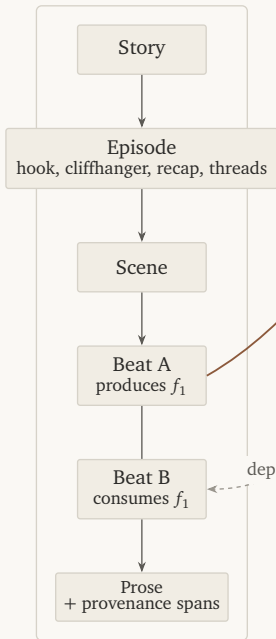
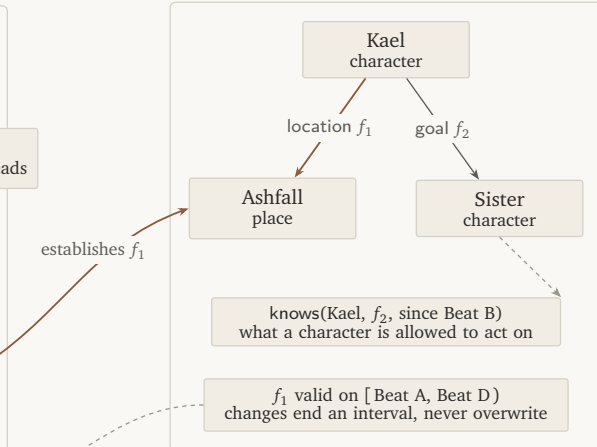


plan tree



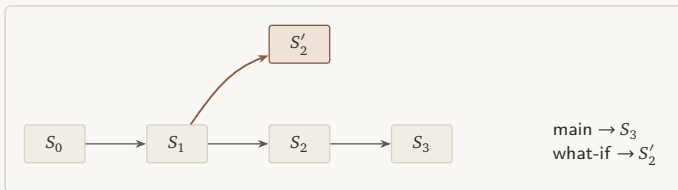
world graph, facts valid over beat intervals



establishes f_1

dependency edge

immutable snapshots, content-addressed and structurally shared



A snapshot is a manifest of content hashes, so unchanged objects are shared between versions. A branch is a pointer; the diff between two snapshots is a set comparison over their manifests, and propagation walks the dependency edges above it.